BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 008 · Primary Storage: RAM, ROM and Cache

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why use lossless compression for source code?

Q2. A bitmap is 40 × 20 pixels, at 4 bits per pixel. What is its raw size in bytes?
A. 800 bytes
B. 3,200 bytes
C. 64 bytes
D. 400 bytes

Q3. How does a bitmap represent pixel colours?

Q4. A microphone records digital sound. “Sampling stores every instant of the sound wave.” Correct this. How does it affect recorded detail?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

RAM — Volatile working memory for programs and data in use.

ROM — Non-volatile memory typically containing firmware.

cache — Small, very fast memory used by the CPU.

volatile — Loses stored contents when power is removed.

New learning

Why is memory needed?: Running programs need fast access to instructions and working data. Primary storage supports the CPU while programs run.

RAM: Random Access Memory stores programs and data currently in use. It is volatile, so its contents are lost when power is removed.

ROM: Read Only Memory is non-volatile and commonly stores firmware or start-up instructions.

Cache: CPU cache is small, very fast memory close to the processor holding frequently used data and instructions, reducing access time.

Worked example

A laptop starts, opens a game, saves a level, then powers off. Explain what each type of memory does.

1. Non-volatile ROM/firmware supplies start-up instructions.

2. The active operating system and game data are loaded into RAM from secondary storage.

3. The processor can access frequently needed data faster from cache.

4. Saving writes the level to non-volatile secondary storage; RAM contents are lost at power-off.

We Do · Practise with guidance

Guided 1

A word processor is open. Where is its working data held?

Worked method: RAM.

Guided 2

Where might the instructions needed to start a device be stored?

Worked method: ROM or another form of non-volatile firmware storage.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Lossless compression restores the original exactly. Losing source-code characters could break the program.

Q2. D — 400 bytes

Q3. Each pixel has a colour code stored in binary.

Q4. Measure amplitude at regular intervals. Store each rounded (quantised) value in binary. Sample rate and bit depth limit the detail captured.

I Do / We Do pages · Guided practice

Q1. RAM.

Q2. ROM or another form of non-volatile firmware storage.

Q3. Cache is much faster and can hold frequently needed instructions/data, reducing time accessing RAM.

You Do pages · Code First

```
score = 10
score = 15
print(score)
```

Expected result: 15

You Do pages · Independent questions

Q1. RAM is volatile [1], while ROM is non-volatile [1].

Q2. Holds frequently used instructions or data near the CPU [1], enabling faster access than repeatedly fetching from RAM [1].

Q3. Saved file persists on non-volatile secondary storage [1]. RAM is volatile [1] and holds working data while the laptop operates [1]. Unsaved edits may be lost when RAM contents disappear at power-off [1].

Exit Ticket · Model answer

False

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Guided 3

Why does the CPU use cache even though the computer has RAM?

Worked method: Cache is much faster and can hold frequently needed instructions/data, reducing time accessing RAM.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the printed value.

```
score = 10
score = 15
print(score)
```

predict

Question 1

State one difference between RAM and ROM. [2 marks]

Question 2

Explain the purpose of cache memory. [2 marks]

Question 3

A pupil saves a document, switches off the laptop and turns it on again. Explain why the document remains but unsaved work could be lost. [4 marks]

Exit Ticket · Answer before checking

RAM usually retains its contents when the power is off.